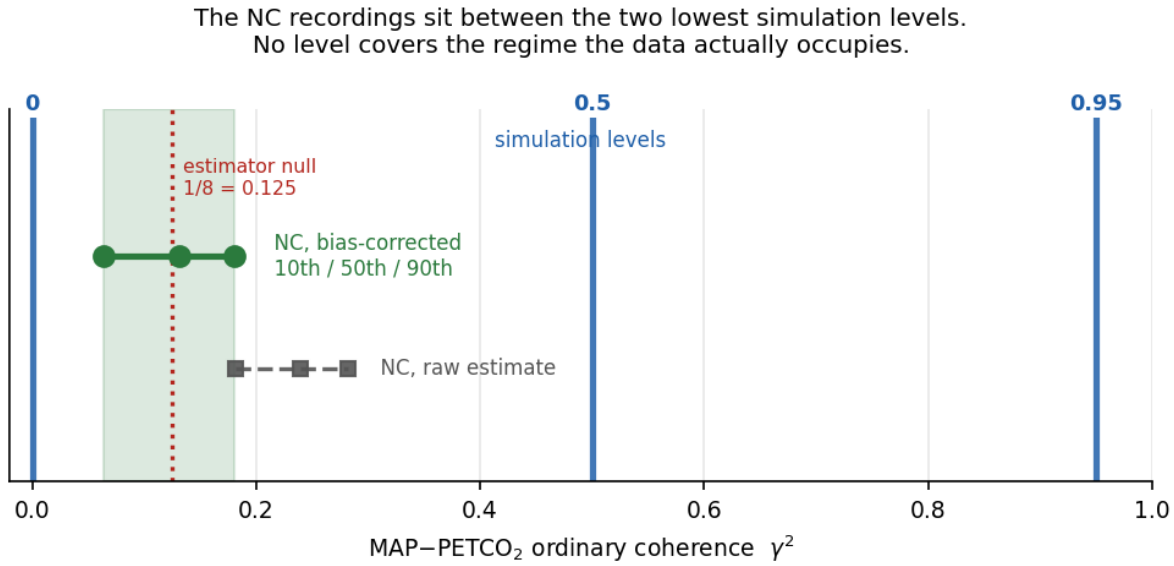
Steps 2 & 3 reviewed

Step 2 execution was clean and step 3's main figure is close to publication quality. But step 2 returned a number that changes what the paper is about, and the pilot contains a result nobody has named yet.

The NC recordings operate at a coherence the simulation does not cover.

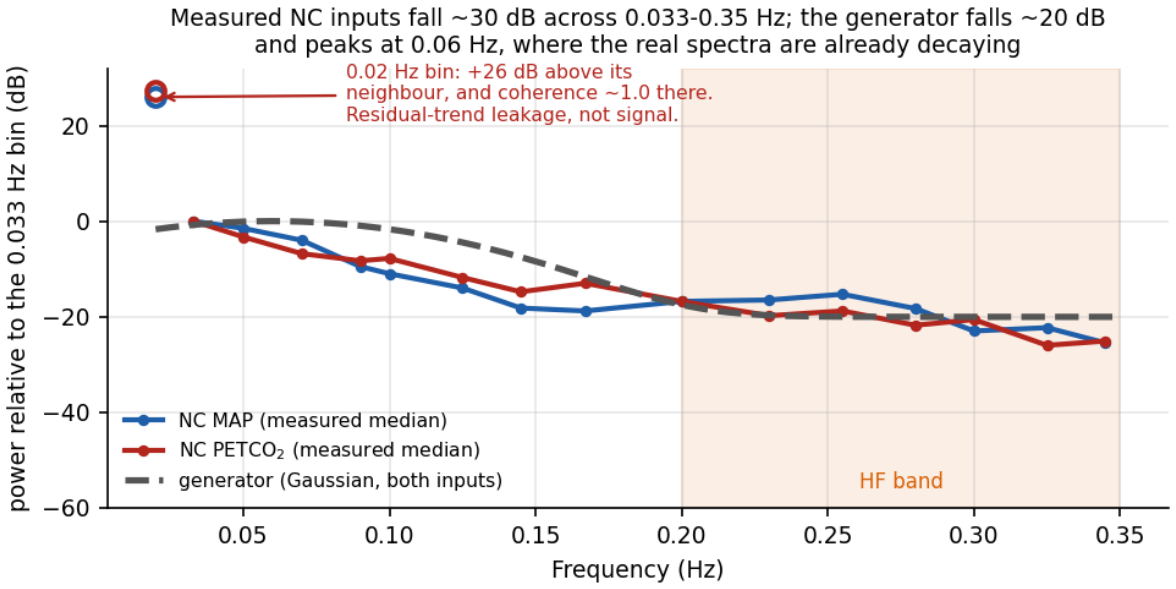
Measured band-mean MAP–$PETCO_2$ coherence is 0.24. The estimator's null expectation at 8 effective averages is 0.125, so more than half of that is bias. Bias-corrected, the 23 subjects span $\gamma^2 \approx$ **0.06–0.18**. The simulation's levels are 0, 0.50 and 0.95. The 10th-percentile subject's raw coherence sits *below* the CARNet threshold, and the median barely clears it.



The simulation's `targetCoherence` is a population parameter; the NC values are sample magnitude-squared coherences from 8 averages, for which the expected estimate is approximately $\gamma^2 + (1-\gamma^2)/n$. They are not comparable until corrected.

Three level decisions, now closed

Deliverable	Measured	FROZEN decision
1. Coherence	raw 0.181 / 0.240 / 0.282 corrected **0.064 / 0.131 / 0.180**	Levels become **[0, 0.15, 0.50, 0.95]**. The added 0.15 is the NC operating point. Core grid goes 9 → 12 scenarios. Without it you cannot answer what the study predicts for the recordings you actually have.
2. Phase lag	–6.17 / +0.27 / +3.68 s, median fit R^2 0.72	Levels become **[–6, 0, +4] s**, provenance upgraded from stress set to **NC-derived**. I checked the spread is real: fitted-lag SE is 0.24–0.72 s against a between-subject SD of ~3.5 s.
3. Input spectra	NC falls ~30 dB over 0.033–0.35 Hz, monotonic	**Reshape required — the freeze trigger has fired.** Replace the Gaussian with a monotonic decay fitted to the NC median. Keep `spectralSimilarity = 1`: MAP and $PETCO_2$ run parallel, so step 2 supports it.



The generator peaks at 0.06 Hz, where the real spectra are already 5–10 dB down and still falling. A steeper roll-off will make HF in-band SNR worse than the pilot shows — see the confound below.

A defect in the lowest bin

MAP–$PETCO_2$ coherence at the lowest retained bin is ≈ **1.0**, falling to 0.33 at the next bin, and both spectra sit ~26 dB above their neighbour there. Coherence of 1.0 between two physiological signals at the bin adjacent to DC is the signature of a shared residual linear trend. The cause is in `estimateWelchSpectra`: **there is no per-segment detrending** — the only option is global and it is off by default. This bin currently anchors the lowest reported band and feeds the band-mean coherence that sets the level above, so the percentiles must be recomputed after the fix.

The pilot contains a second mechanism, and it is the one that matters for NC

In the coherence panel, row $\gamma^2 = 0$, MISO is **better** — not neutral. At zero coherence there is no omitted-variable bias, so SISO is unbiased. MISO still wins because including CO_2 removes CO_2-driven variance from the unexplained CBFV, lowering the variance of the MAP coefficient.

Mechanism	Requires input coherence?	Where it appears
Removes omitted-input **bias**	Yes — scales with $\sqrt{\gamma^2}$	γ^2 = 0.5 and 0.95 rows, low frequency
Reduces residual **variance**	**No**	present even at $\gamma^2 = 0$

> **Why this is the paper's spine**
>
> Your NC data sits at $\gamma^2 \approx 0.15$, where the bias mechanism is nearly absent. So the predicted NC advantage comes almost entirely from variance reduction: small, positive, concentrated at low frequency. That is a specific falsifiable prediction you can check against the NC bootstrap result you already have. If it matches, it validates the whole framework. This framing currently exists nowhere in the project.

Figure review

Page 1 works. Colour limits are symmetric and identical across all four panels — the thing most people get wrong — and the bias–variance crossover at $\gamma^2 = 0.95$ is visible without being told to look for it. The two companion pages need work:

Page	Problem	Fix
2 — analytic reference	Plots raw SISO bias, not predicted *advantage*: different units, scale and colormap from page 1, so the two cannot be compared. Colour range is consumed by the single 0.02 Hz column. Duration and SNR panels are constant by construction — a population quantity cannot depend on record length or noise.	Two panels only (coherence, pathway ratio), in $\log_{10}$ advantage on page 1's colour scale. Add a measured-minus-predicted difference map — that is the actual scientific content.
3 — realised SNR	Did its job: confirms SNR falls ~25 dB at 0.02 Hz to ~0 dB at 0.30–0.35 Hz. But panel 4's scale runs 0–160 dB because the noiseless row is infinite, compressing the informative rows into one colour.	Exclude Inf from the colour scaling; label the noiseless row as a separate category.
all	No N annotation; grey-cell convention never explained; NC operating range not marked; captions do not state the estimand.	All four are what a physiologist needs to read the figure, and none are present.

The one thing that must change before the paper run

The frequency axis confounds frequency with in-band SNR. Page 3 shows realised SNR falling ~25 dB across the band. Page 1 shows MISO losing to SISO at high frequency when $\gamma^2 = 0.95$. Two mechanisms produce that orange region — collinearity, and a two-parameter estimator degrading faster as local SNR approaches 0 dB — and the current design cannot separate them. After the spectral reshape the gradient gets steeper and the problem worsens.

> **Fix**
>
> Add `settings.observations.noiseSpectrum` with `"white"` (current) and `"signalShaped"` (noise follows the clean CBFV spectrum, so in-band SNR is flat and "10 dB" means 10 dB at every frequency). Run the primary factorial with `signalShaped`; keep `white` as a named robustness contrast. Then any frequency dependence in the advantage map is attributable to pathway shape and coherence — which is what the figure design was for.

Step 4, in order

#	Task	Note
1	Per-segment linear detrending in `estimateWelchSpectra`, then re-run step 2	Prerequisite
2	Recompute the bias-corrected coherence percentiles; confirm the 0.15 level	Prerequisite
3	Update frozen levels: coherence [0, 0.15, 0.50, 0.95]; phase [–6, 0, +4] s	Prerequisite
4	Reshape input spectra to a monotonic decay fitted to the NC median	Changes generated data
5	Add `noiseSpectrum`; make `signalShaped` primary	Changes generated data
6	Rebuild the analytic reference as predicted advantage + difference map	Presentation
7	Fix the figure defects (N, grey cells, NC range, estimand in captions)	Presentation
8	Two new figures: predicted advantage at γ^2=0.15 vs measured NC; minimum detectable advantage	Presentation
9	Write the two-mechanism framing into the protocol document	Documentation

Items 1–3 land together. Items 4–5 force a re-run, so do them before the production run, not after. Items 6–9 are presentation and can proceed in parallel. Four new acceptance tests are listed in §E of the full review.

Not verified: MATLAB did not run — all numbers come from independent re-implementation. I did not see the exported tables, so the coherence and lag values are as reported in `STEP2_NC_INPUT_AUDIT.md`, and the n = 8 effective averages is inferred from the CARNet threshold shown in the figure. NC spectral values were read off the figure, so the reshape fit must use the exported table, not

mine. The lowest-bin diagnosis is a strong inference from three converging signs, not a direct measurement — task 1 will confirm or refute it.